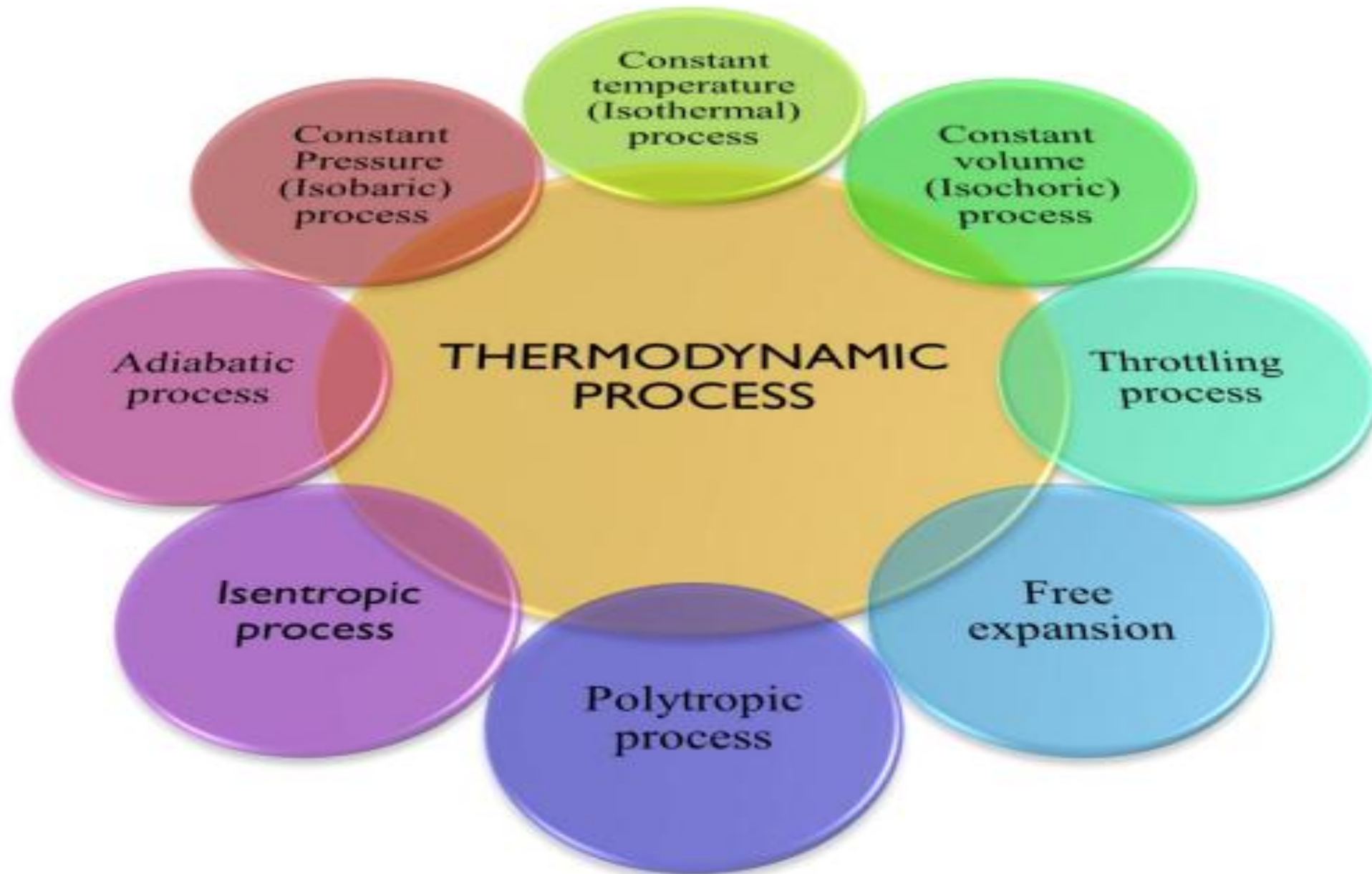




Equation Used

- $W_{1-2} = \int_1^2 P dV$
- $dU = mc_v(T_2 - T_1)$ or $U_2 - U_1 = mc_v(T_2 - T_1)$
- $\delta Q - \delta W = dU$ or $Q_{1-2} = dU + W_{1-2}$
- $dH = mc_p(T_2 - T_1)$ or $H_2 - H_1 = mc_p(T_2 - T_1)$

1. Constant Volume(Isochoric) Process

Property Relationship

$$p_1 V_1 = mRT_1$$

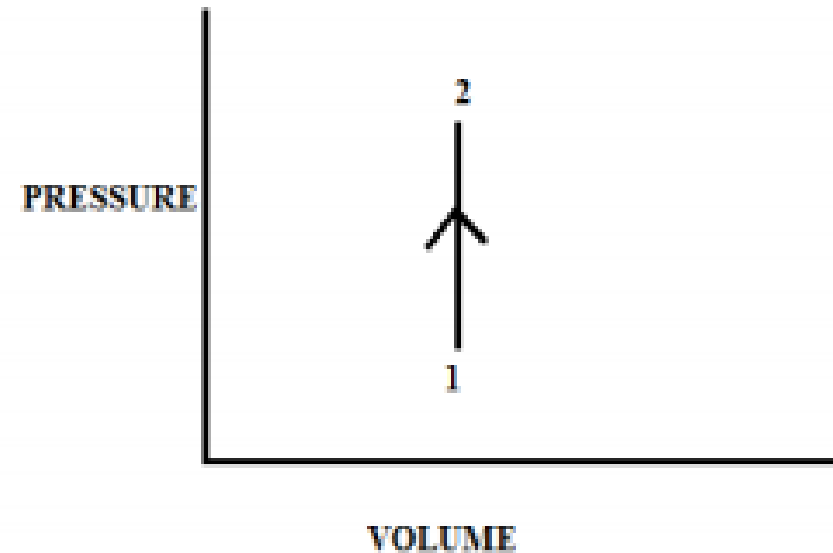
and

$$p_2 V_2 = mRT_2$$

Since $V_1 = V_2$, by dividing one by the other we can write,

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

P – V diagram



Constant Volume(Isochoric) Process

Work done during the process

Work done can be obtained from the area under the curve in the p-V diagram.

$$W = \int_1^2 p \, dV$$

Since $dV = 0$,

$$W = 0$$

i.e., in a constant volume process, no displacement work is done.

Change in internal energy

For an ideal gas, change in internal energy per unit mass is given by the expression,

$$\begin{aligned}\Delta u &= \int_1^2 C_v \, dT \\ &= C_v (T_2 - T_1)\end{aligned}$$

For a system of mass 'm', we have,

$$\Delta U = m C_v (T_2 - T_1)$$

Constant Volume(Isochoric) Process

Heat transferred

The general equation for heat transferred per unit mass is,

$$q = \int_1^2 C dT$$

For a constant volume process, it is

$$q = C_v (T_2 - T_1) = \Delta u$$

For a system of mass 'm', we can write the total heat transferred as,

$$Q = m C_v (T_2 - T_1)$$

i.e., for a constant volume process, change in internal energy is always equal to the heat transferred during the process.

Change in enthalpy

For an ideal gas, change in enthalpy is given by,

$$\Delta h = \int_1^2 C_p dT$$
$$C_p (T_2 - T_1)$$

For a system of mass 'm', the change in total enthalpy is,

$$\Delta H = m C_p (T_2 - T_1)$$

2. Constant Pressure(Isobaric) Process

Property Relationship

$$p_1 V_1 = m R T_1$$

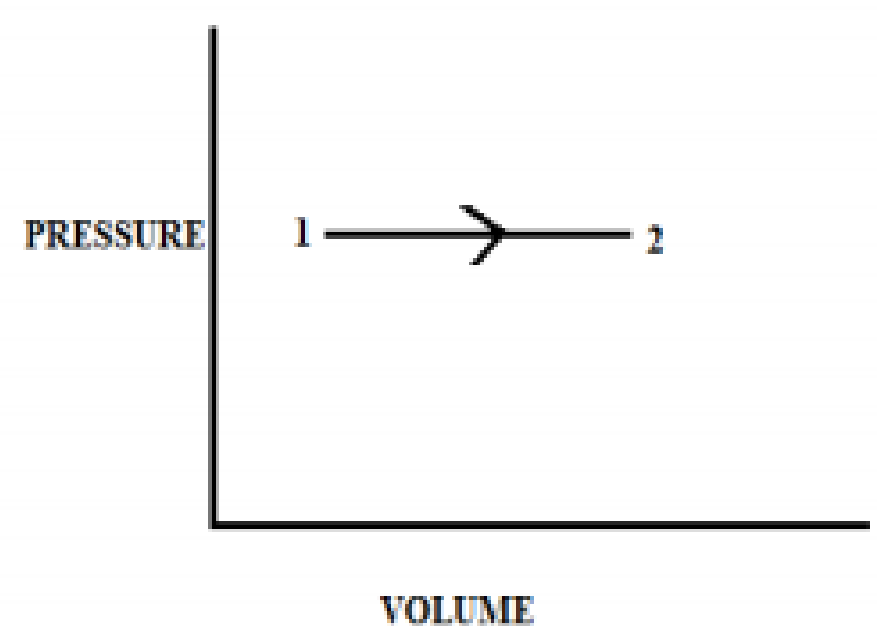
and

$$p_2 V_2 = m R T_2$$

Dividing one by the other, and since $p_1 = p_2$, we get,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

P – V diagram



Constant Pressure(Isobaric) Process

⇒ Work done during the process

The non-flow work done by the process is the area under the curve in p-V diagram

$$\begin{aligned} W &= \int_1^2 p dV = p \int_1^2 dV \\ &= p (V_2 - V_1) \\ &= m R (T_2 - T_1) \quad (\because pV = mRT) \end{aligned}$$

⇒ Change in internal energy

For an ideal gas, change in internal energy per unit mass is,

$$\begin{aligned} \Delta u &= \int_1^2 C_v dT \\ &= C_v (T_2 - T_1) \end{aligned}$$

For a given mass m, the change in total internal energy is,

$$\Delta U = m C_v (T_2 - T_1)$$

Constant Pressure(Isobaric) Process

⇒ Heat transferred

The general equation for heat transferred per unit mass is,

$$q = \int_1^2 C dT$$
$$= C_p (T_2 - T_1)$$

For a given system of mass m , the heat transferred is,

$$Q = m C_p (T_2 - T_1)$$

⇒ Change in enthalpy

For an ideal gas, change in enthalpy is,

$$\Delta h = \int_1^2 C_p dT$$
$$= C_p (T_2 - T_1)$$

For a system of mass m , the change in the total enthalpy is,

$$\Delta H = m C_p (T_2 - T_1)$$

For a constant pressure process, the change in enthalpy is equal to the heat added during the process.

3. Constant Temperature(Isothermal) Process

Property Relationship

$$p_1 V_1 = m R T_1$$

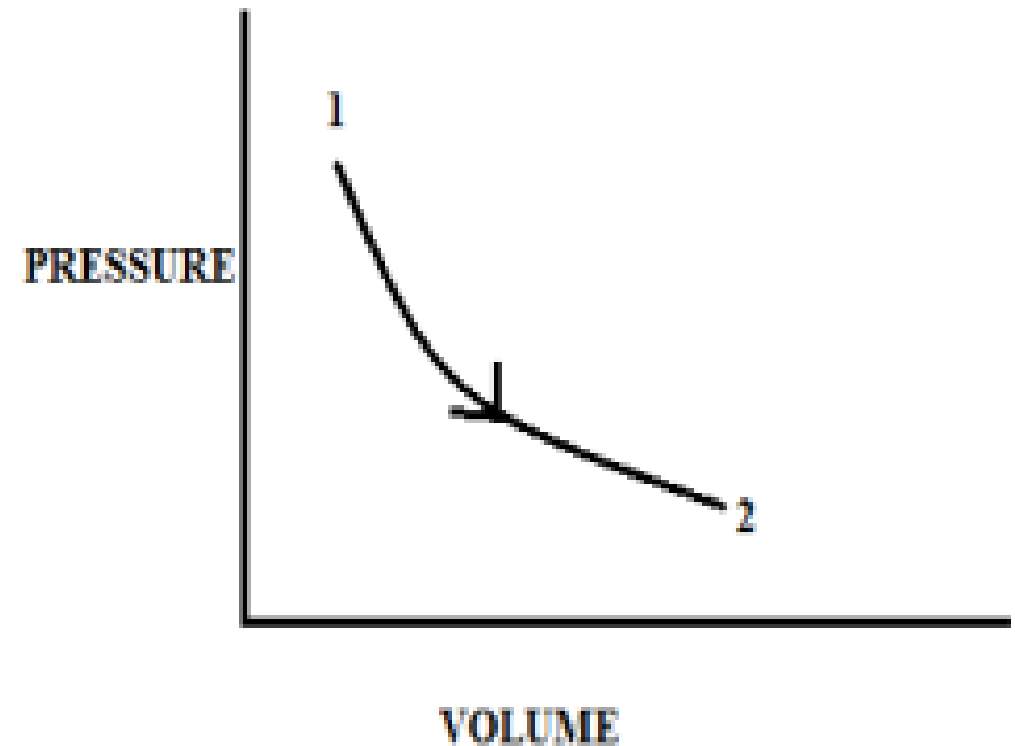
and $p_2 V_2 = m R T_2$

Since $T_1 = T_2$ for an isothermal process, we get

$$p_1 V_1 = p_2 V_2$$

or $pV \text{ constant}$

P – V diagram



Constant Temperature(Isothermal) Process

⇒ Work done

For isothermal process we have, $pV = \text{constant (C)}$. Therefore, we have

$$p = \frac{C}{V}$$

Substituting it in the general expression for work, we get

$$\begin{aligned} W &= \int_1^2 p \, dV = \int_1^2 C \frac{dV}{V} \\ &= C \ln \frac{V_2}{V_1} \end{aligned}$$

But $C = p_1 V_1 = p_2 V_2$. Substituting it in the expression for work, we get,

$$W = p_1 V_1 \ln \frac{V_2}{V_1}$$

Also, since $pV = mRT$, for a perfect gas, we can write,

$$W = mRT_1 \ln \frac{V_2}{V_1}$$

Also, since $p_1 V_1 = p_2 V_2$, we can also write the expression for work as,

$$W = p_1 V_1 \ln \frac{p_1}{p_2} = mRT_1 \ln \frac{p_1}{p_2}$$

Constant Temperature(Isothermal) Process

➡ Change in internal energy

For an ideal gas change in internal energy is given by,

$$\begin{aligned}\Delta u &= \int_1^2 C_v dT \\ &= C_v (T_2 - T_1) = 0\end{aligned}$$

i.e., for an isothermal process, the change in internal energy is zero.

➡ Heat transferred

From first law of thermodynamics, we

$$Q = \Delta U + W$$

Since $\Delta U = 0$ for an isothermal process,

$$Q = W = p_1 V_1 \ln \frac{V_2}{V_1} = m R T_1 \ln \frac{V_2}{V_1}$$

$$= p_1 V_1 \ln \frac{p_1}{p_2} = m R T_1 \ln \frac{p_1}{p_2}$$

i.e., for an isothermal process, heat transferred is equal to work done.

Constant Temperature(Isothermal) Process

Change in enthalpy

For an ideal gas, change in enthalpy is

$$\Delta h = \int_1^2 C_p dT = C_p (T_2 - T_1) = 0$$

i.e., for an isothermal process, the change in enthalpy of the gas is zero.

Numerical Problems:-

1.) A rigid tank is having a capacity of 1 m^3 . It contains nitrogen gas at a pressure and temperature of 0.5 Mpa and 30°C . 250 kJ of heat is added to the gas. Determine:

(i) Final pressure and temperature

(ii) Change in internal energy, Take $\gamma = 1.4$ and $C_v = 743$
J/kg K

Given :

$$V_1 = V_2 = 1 \text{ m}^3$$

$$p_1 = 0.5 \text{ Mpa}$$

$$T_1 = 30^\circ\text{C} = 303 \text{ K}$$

$$Q = 250 \text{ kJ}$$

For nitrogen gas,

$$C_p/C_v = 1.4 \text{ and } C_v = 743 \text{ kJ/kgK.}$$

$$\therefore C_p = 1.4 \times 743 = 1040.2 \text{ J/kgK.}$$

$$R = C_p - C_v = 297.2 \text{ J/kgK.}$$

i) Final pressure and temperature

Using the characteristic gas equation,

$$m = \frac{p_1 V_1}{RT_1} = \frac{0.5 \times 10^6 \times 1}{297.2 \times 303} = \underline{5.552 \text{ kg}}$$

Final temperature can be calculated from the equation for heat added. For constant volume process,

$$Q = m C_v (T_2 - T_1)$$

$$250 \times 10^3 = 5.552 \times 743 \times (T_2 - 303)$$

$$\therefore \underline{T_2 = 363.6 \text{ K} = 90.6^\circ\text{C.}}$$

$$\text{Final pressure, } p_2 = p_1 \times \frac{T_2}{T_1} = 0.5 \times \frac{363.6}{303} = \underline{0.6 \text{ MPa}}$$

ii) Change in internal energy

$$\Delta U = Q$$

$$= \underline{250 \text{ kJ}}$$

($\because$ constant volume process)

2). A gas is compressed in a cylinder from a volume of 0.07 m^3 to a final volume of 0.055 m^3 . The initial pressure and temperature of the gas is 150 kPa and 100°C .

If the compression is carried out at constant pressure, find

(i) temperature at the end of compression

(ii) Work spent in compressing the gas

(iii) Heat transfer from the gas

(iv) Change in internal energy

Assume $C_p = 1.005 \text{ kJ/kgK}$ and $R = 285 \text{ J/kgK}$

Given :

$$V_1 = 0.07 \text{ m}^3$$

$$p_1 = 150 \text{ kPa}$$

$$T_1 = 100^\circ\text{C} = 373 \text{ K}$$

$$Q = 250 \text{ kJ}$$

$$V_2 = 0.055 \text{ m}^3$$

$$C_p = 1.005 \text{ kJ/kgK} \text{ and } R = 285 \text{ J/kgK}$$

i) For a constant pressure process, V/T is constant.

$$\therefore \underline{T_2} = \frac{V_2}{V_1} \times T_1 = \frac{0.055}{0.07} \times 373$$
$$= \underline{293.07 \text{ K} = 20.07^\circ\text{C}}$$

ii) Work done

$$W = p (V_2 - V_1) = 150 \times 10^3 \times (0.055 - 0.07)$$
$$= \underline{-2.25 \times 10^3 \text{ J} = -2.25 \text{ kJ}}$$

Since work is done on the gas, it is negative

iii) Heat transferred

Mass of gas compressed,

$$m = \frac{p_1 V_1}{RT_1} = \frac{150 \times 10^3 \times 0.07}{285 \times 373} = \underline{0.09877 \text{ kg}}$$

Heat transferred

$$\begin{aligned} Q &= m C_p (T_2 - T_1) = 0.09877 \times 1.005 \times (293.07 - 373) \\ &= \underline{-7.93 \text{ kJ}} \end{aligned}$$

Since heat is transferred from the gas, it is negative.

iv) Change in internal energy

From the perfect gas relation

$$C_v = C_p - R = 1005 - 285 = 720 \text{ J/kg K}$$

$$\begin{aligned} \Delta U &= m C_v (T_2 - T_1) = 0.09877 \times 0.72 (293.07 - 373) \\ &= \underline{-5.684 \text{ kJ}} \end{aligned}$$

There is a decrease in internal energy

Reversible Adiabatic(Isentropic) Process

$$\underline{\delta Q = du + \delta W}$$

Since for an adiabatic process, $\delta Q = 0$, we get

$$du + \delta W = 0$$

For a perfect gas, we have $du = C_v dT$ and $\delta W = p dv$. So, the equation may be written as,

$$C_v dT + p dv = 0$$

For unit mass of perfect gas, we have

$$pv = RT$$

Differentiating both sides of the expression, we get

$$p dv + v dp = R dT \quad (R \text{ is constant})$$

$$\text{or} \quad dT = \frac{p dv + v dp}{R}$$

Substituting for dT in Eq.

$$C_v \left(\frac{p dv + v dp}{R} \right) + p dv = 0$$

$$\underline{C_v p dv + C_v v dp + R p dv = 0}$$

Reversible Adiabatic(Isentropic) Process

$$C_v p dv + C_v v dp + R p dv = 0$$

For an ideal gas, $R = C_p - C_v$. Substituting in the equation, we get

$$C_v p dv + C_v v dp + (C_p - C_v) p dv = 0$$

$$\text{or } C_v v dp + C_p p dv = 0$$

Dividing both sides by $C_v p v$, we get

$$\frac{dp}{p} + \frac{C_p}{C_v} \frac{dv}{v} = 0$$

For any gas, the ratio of the two specific heats (C_p and C_v) is defined as **adiabatic index** denoted by γ . Substituting adiabatic index in the equation, we get,

$$\frac{dp}{p} + \gamma \frac{dv}{v} = 0$$

Integrating the above equation, we get

$$\ln p + \gamma \ln v = \text{constant}$$

Taking antilog on both sides, the equation becomes

$$p v^\gamma = C$$

Reversible Adiabatic(Isentropic) Process

Property relationship

From the ideal gas equation, $p_1 V_1 = m R T_1$ and $p_2 V_2 = m R T_2$, we can write

$$\frac{T_2}{T_1} = \frac{p_2 V_2}{p_1 V_1} \quad \text{--- (1)}$$

For an isentropic process, $pV^\gamma = \text{constant}$. Therefore,

$$p_1 V_1^\gamma = p_2 V_2^\gamma$$

$$\text{or } \frac{p_2}{p_1} = \left(\frac{V_1}{V_2} \right)^\gamma = \left(\frac{V_2}{V_1} \right)^{-\gamma} \quad \text{--- (2)}$$

$$\text{or } \frac{V_2}{V_1} = \left(\frac{p_2}{p_1} \right)^{-\frac{1}{\gamma}} \quad \text{--- (3)}$$

Substituting Eq. (2) in (1), we get

$$\frac{T_2}{T_1} = \left(\frac{V_2}{V_1} \right)^{-\gamma} \left(\frac{V_2}{V_1} \right) = \left(\frac{V_2}{V_1} \right)^{1-\gamma}$$

Substituting Eq. (3) in Eq. (1), we get

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right) \left(\frac{p_2}{p_1} \right)^{-\frac{1}{\gamma}} = \left(\frac{p_2}{p_1} \right)^{\left(\gamma - \frac{1}{\gamma} \right)}$$

Reversible Adiabatic(Isentropic) Process

Work done

From the law for an isentropic process, we have

$$pV^\gamma = C \quad \text{or} \quad p = \frac{C}{V^\gamma}$$

Substituting it in the expression for work done, we get

$$W = \int_1^2 p dV = \int_1^2 \frac{C}{V^\gamma} dV$$

Integrating, we get

$$W = C \left[\frac{V^{-\gamma+1}}{-\gamma+1} \right]_{V_1}^{V_2} = C \left[\frac{V_2^{-\gamma+1} - V_1^{-\gamma+1}}{1-\gamma} \right]$$

Since $C = p_1 V_1^\gamma = p_2 V_2^\gamma$, the equation becomes,

$$W = \frac{p_2 V_2^{1-\gamma} - p_1 V_1^{1-\gamma}}{1-\gamma}$$

$$= \frac{p_2 V_2 - p_1 V_1}{1-\gamma}$$

Reversible Adiabatic(Isentropic) Process

$$= \frac{p_2 V_2 - p_1 V_1}{1 - \gamma}$$

Also, since $pV = mRT$, the equation for work done can also be written as,

$$W = \frac{m R (T_2 - T_1)}{1 - \gamma}$$

Substituting $\gamma = C_p/C_v$ and $R = C_p - C_v$, in Eq. we get

$$W = \frac{m(C_p - C_v)(T_2 - T_1)}{1 - \frac{C_p}{C_v}} = \frac{m(C_p - C_v)(T_2 - T_1)}{\frac{C_v - C_p}{C_v}}$$

or $W = - m C_v (T_2 - T_1)$

i.e., during an isentropic process, since no heat is added to the system, work is done by the system by extracting its internal energy

Reversible Adiabatic(Isentropic) Process

Change in internal energy

From the expression for change in internal energy,

$$\Delta U = \int_1^2 m C_v dT = m C_v (T_2 - T_1) = -W$$

Heat transferred

Since the process is adiabatic, the heat transferred from or to the system is zero

$$Q = 0$$

Change in enthalpy

From the general equation for change in enthalpy of a perfect gas

$$\Delta h = \int_1^2 C_p dT = C_p (T_2 - T_1)$$

Polytropic Process

- It is the general thermodynamic process, obtained by generalising all thermodynamic processes

$$PV^n = \text{constant}$$

$$P_1 V_1^n = P_2 V_2^n = P_3 V_3^n = PV^n = \text{constant}$$

Polytropic Process

Property relationship

Property relations for a polytropic process can be obtained by replacing the adiabatic index γ by the polytropic index n

$$\frac{p_2}{p_1} = \left(\frac{V_2}{V_1} \right)^{-n} = \left(\frac{V_1}{V_2} \right)^n$$

$$\frac{V_2}{V_1} = \left(\frac{p_2}{p_1} \right)^{-\frac{1}{n}} = \left(\frac{p_1}{p_2} \right)^{\frac{1}{n}}$$

$$\frac{T_2}{T_1} = \left(\frac{V_2}{V_1} \right)^{1-n}$$

$$\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{\frac{(n-1)}{n}}$$

Polytropic Process

Heat exchanged.

From first law,

$$\underline{{}_1Q_2 = \Delta U + {}_1W_2}$$

$$\text{ie., } {}_1Q_2 = mC_v (T_2 - T_1) + \frac{mR}{(n-1)} (T_1 - T_2)$$

$$= m(T_1 - T_2) \left\{ \frac{R}{(n-1)} - C_v \right\}$$

$$C_p - C_v = R$$

Dividing by C_v

$$\frac{C_p}{C_v} - 1 = \frac{R}{C_v}$$

$$\gamma - 1 = \frac{R}{C_v}$$

$$\text{or } C_v = \frac{R}{(\gamma - 1)}$$

$$\begin{aligned}
 {}_1Q_2 &= m(T_1 - T_2) \left\{ \frac{R}{(n-1)} - \frac{R}{(\gamma-1)} \right\} \\
 &= mR(T_1 - T_2) \left\{ \frac{1}{(n-1)} - \frac{1}{(\gamma-1)} \right\} \\
 &= mR(T_1 - T_2) \left\{ \frac{\gamma-1-n+1}{(n-1)(\gamma-1)} \right\} \\
 &= mR(T_1 - T_2) \left\{ \frac{(\gamma-n)}{(n-1)(\gamma-1)} \right\}
 \end{aligned}$$

$$= \frac{\gamma-n}{\gamma-1} \frac{mR(T_1 - T_2)}{(n-1)}$$

$${}_1Q_2 = \frac{\gamma-n}{\gamma-1} \times \text{Workdone}$$

Polytropic Process

Expression for polytropic index 'n'.

For a polytropic process,

$$p_1 V_1^n = p_2 V_2^n$$

$$\frac{p_1}{p_2} = \left\{ \frac{V_2}{V_1} \right\}^n$$

Taking logarithm,

$$\ln \left\{ \frac{p_1}{p_2} \right\} = n \ln \left\{ \frac{V_2}{V_1} \right\}$$

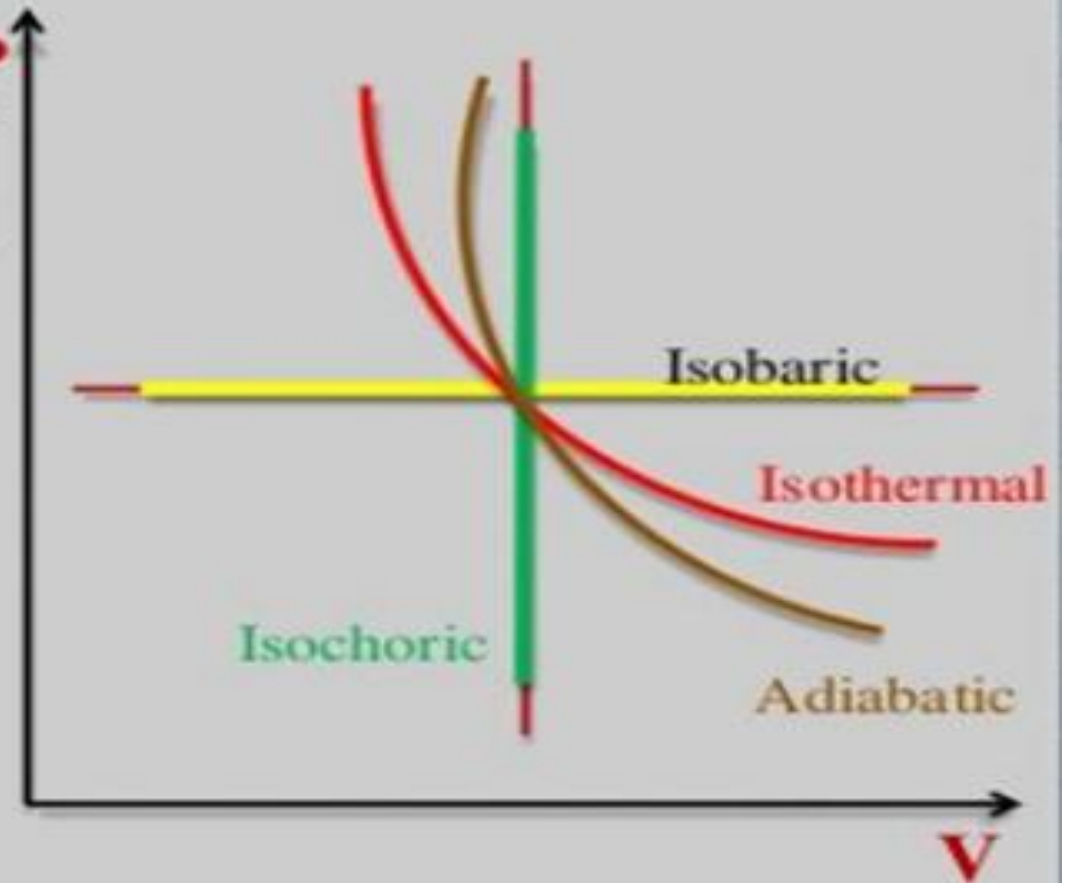
$$\text{or } n = \frac{\ln \left(\frac{p_1}{p_2} \right)}{\ln \left(\frac{V_2}{V_1} \right)}$$

Polytropic process for various values of 'n'

Gas Process

$PV^n = C \rightarrow$ General (polytropic) Process

Value of n	$PV^n = C$	Process law	Process name
$n = 0$	$PV^0 = C$	$P = C$	Isobaric
$n = 1$	$PV^1 = C$	$PV = C$	Isothermal
$n = \gamma$	$PV^\gamma = C$	$PV^\gamma = C$	Adiabatic
$n = \infty$	$PV^\infty = C$	$V = C$	Isochoric



Summary of processes for perfect gas

Process	Index n	Heat added	$\int_1^2 p dv$	p, v, T relations	Specific heat, c
Constant pressure	$n = 0$	$c_p(T_2 - T_1)$	$p(v_2 - v_1)$	$\frac{T_2}{T_1} = \frac{v_2}{v_1}$	c_p
Constant volume	$n = \infty$	$c_v(T_2 - T_1)$	0	$\frac{T_1}{T_2} = \frac{p_1}{p_2}$	c_v
Constant temperature	$n = 1$	$p_1 v_1 \log_e \frac{v_2}{v_1}$	$p_1 v_1 \log_e \frac{v_2}{v_1}$	$p_1 v_1 = p_2 v_2$	∞
Reversible adiabatic	$n = \gamma$	0	$\frac{p_1 v_1 - p_2 v_2}{\gamma - 1}$	$p_1 v_1^\gamma = p_2 v_2^\gamma$ $\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{\gamma-1} = \left(\frac{p_2}{p_1}\right)^{\frac{\gamma-1}{\gamma}}$	0
Polytropic	$n = n$	$c_n(T_2 - T_1)$ $= c_v \left(\frac{\gamma - n}{1 - n}\right) \times (T_2 - T_1)$ $= \frac{\gamma - n}{\gamma - 1} \times \text{work done (non-flow)}$	$\frac{p_1 v_1 - p_2 v_2}{n - 1}$	$p_1 v_1^n = p_2 v_2^n$ $\frac{T_2}{T_1} = \left(\frac{v_1}{v_2}\right)^{n-1} = \left(\frac{p_2}{p_1}\right)^{\frac{n-1}{n}}$	$c_n = c_v \left(\frac{\gamma - n}{1 - n}\right)$

Numerical Problems:-

1 A gas at a pressure and temperature of 1.5 MPa and 300°C is expanded isentropically to a pressure 0.3 MPa. If the initial volume of the gas is 0.07 m³, calculate (i) Mass of gas (ii) Final temperature (iii) Work transfer (iv) change in entropy. Take $C_p = 1.04$ kJ/kgK and $C_v = 0.74$ kJ/kgK.

Solution. Given : $p_1 = 1.5 \text{ MPa}$, $T_1 = 300^\circ\text{C} = 573 \text{ K}$, $p_2 = 0.3 \text{ MPa}$, $V_1 = 0.07 \text{ m}^3$, $C_p = 1.04 \text{ kJ/kg K}$, $C_v = 0.74 \text{ kJ/kg K}$

For the given gas,

$$\gamma = C_p / C_v = 1.405$$

$$R = C_p - C_v = 0.3 \text{ kJ/kg K.}$$

i) From ideal gas equation, $pV = mRT$,

$$m = \frac{p_1 V_1}{RT_1} = \frac{1.5 \times 10^6 \times 0.07}{300 \times 573} = 0.6108 \text{ kg}$$

ii) For isentropic process, we have

$$T_2 = T_1 \times \left(\frac{p_2}{p_1} \right)^{(\gamma-1)/\gamma} = 573 \times \left(\frac{0.3}{1.5} \right)^{0.405/1.405}$$
$$= 360.3 \text{ K} = 87.3^\circ\text{C}$$

iii) Work transfer

$$W = \frac{m R (T_2 - T_1)}{1 - \gamma} = \frac{0.6108 \times 300 (360.3 - 573)}{1 - 1.405}$$
$$= 96235 \text{ J} = 96.235 \text{ kJ}$$

Numerical Problems:-

- 2 A certain quantity of air has a volume of 0.028 m^3 at pressure of 1.25 bar and 25°C . It is compressed to a volume of 0.0042 m^3 according to the law $pV^{1.3}$ constant. Find the final temperature and work done during compression. Also determine the reduction in pressure at a constant volume required to bring the air back to its original temperature.

Solution:

Given:

$$V_1 = 0.028 \text{ m}^3$$

$$V_2 = 0.0042 \text{ m}^3$$

$$n = 1.3$$

$$T_1 = 25^\circ\text{C} = 298 \text{ K}$$

$$T_3 = T_1 = 298 \text{ K}$$

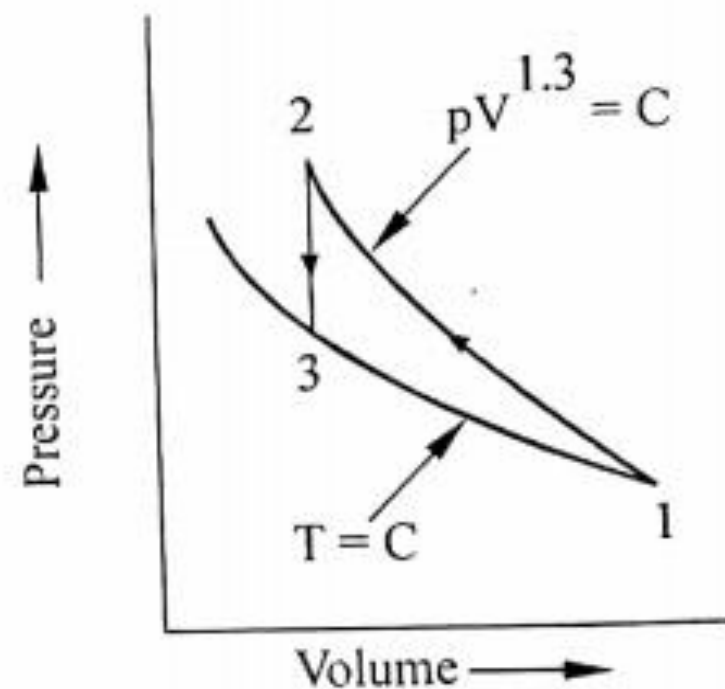
$$p_1 = 1.25 \text{ bar} = 1.25 \times 10^5 \text{ N/m}^2$$

To find: T_2 W ($p_2 - p_3$)

For the polytropic process, 1 - 2,

$$\frac{T_1}{T_2} = \left\{ \frac{V_2}{V_1} \right\}^{n-1}$$

$$T_2 = \frac{T_1}{\left\{ \frac{V_2}{V_1} \right\}^{n-1}}$$



$$T_2 = \frac{298}{\left\{ \frac{0.0042}{0.028} \right\}^{1.3-1}}$$

$$T_2 = 526.49 \text{ K} = 253.49^\circ\text{C}$$

Also, $\frac{p_1}{p_2} = \left\{ \frac{V_2}{V_1} \right\}^n$

$$p_2 = \frac{p_1}{\left\{ \frac{V_2}{V_1} \right\}^n} = \frac{1.25 \times 10^5}{\left\{ \frac{0.0042}{0.028} \right\}^{1.3}}$$

$$= 14.723 \times 10^5 \text{ N/m}^2$$

$${}_1W_2 = \frac{p_1 V_1 - p_2 V_2}{n-1}$$

$$= \frac{1.25 \times 10^5 \times 0.028 - 14.723 \times 10^5 \times 0.0042}{1.3 - 1}$$

$${}_1W_2 = -8.946 \text{ kJ}$$

For the constant volume process 2 - 3,

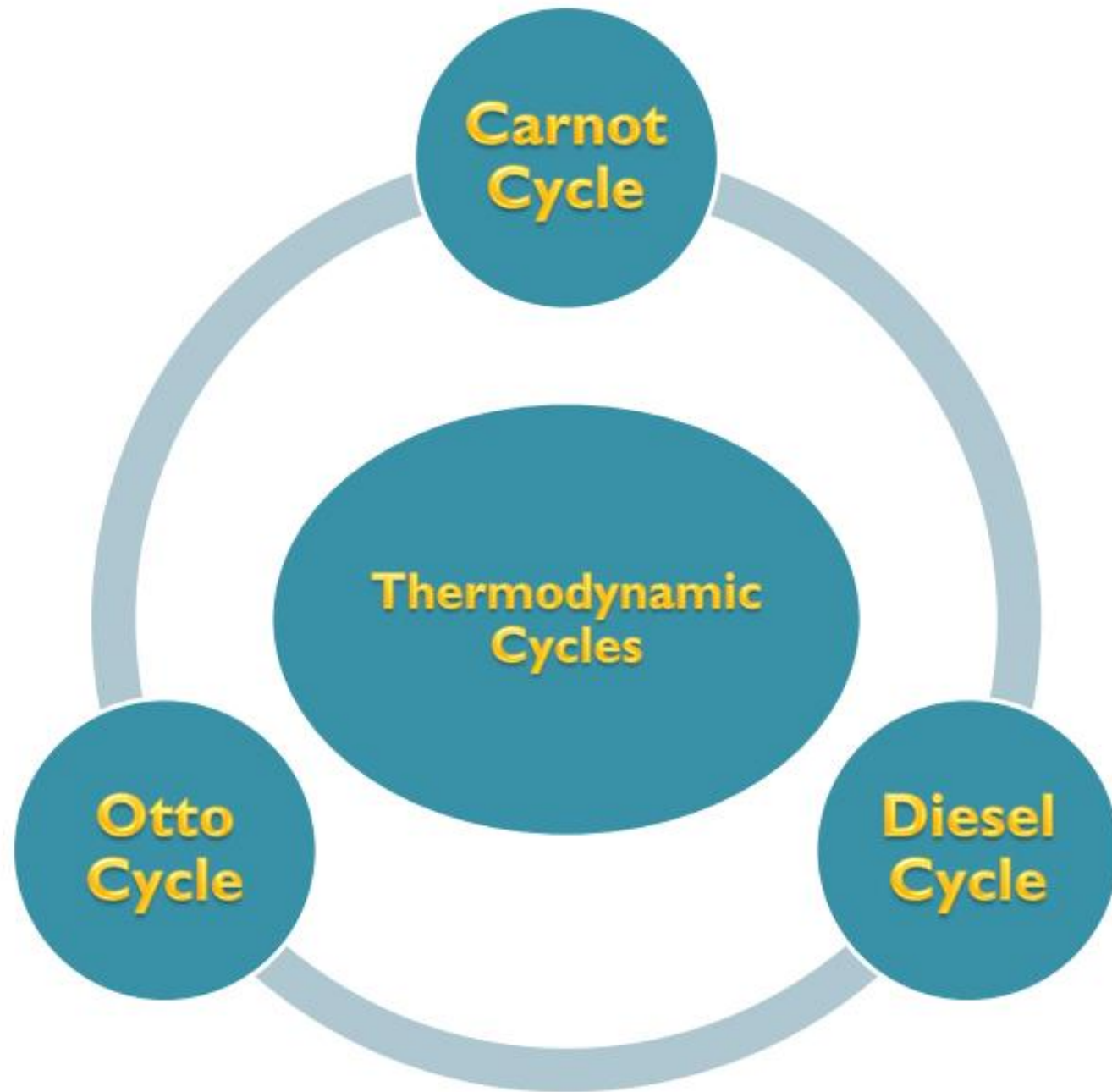
$$\frac{p_3}{T_3} = \frac{p_2}{T_2}$$

$$p_3 = \frac{p_2}{T_2} \times T_3$$

$$p_3 = \frac{14.723 \times 10^5 \times 298}{526.49} = 8.33 \times 10^5 \text{ N/m}^2$$

$$p_2 - p_3 = 14.723 \times 10^5 - 8.33 \times 10^5$$

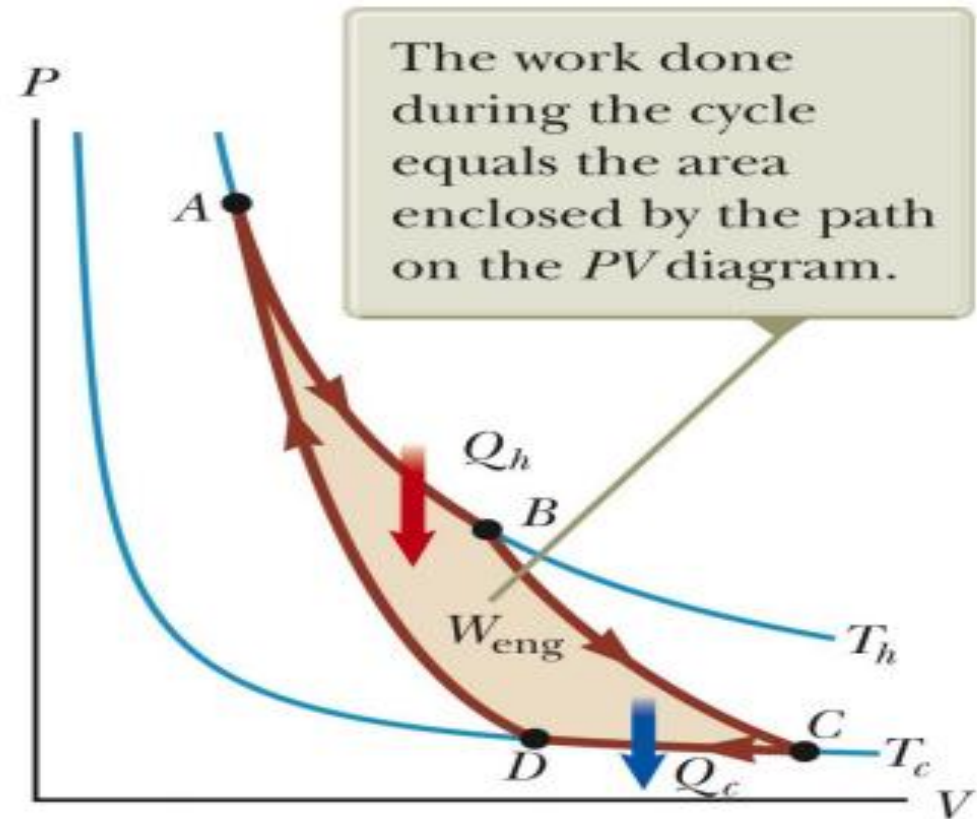
$$\underline{\underline{p_2 - p_3 = 6.393 \text{ bar}}}$$



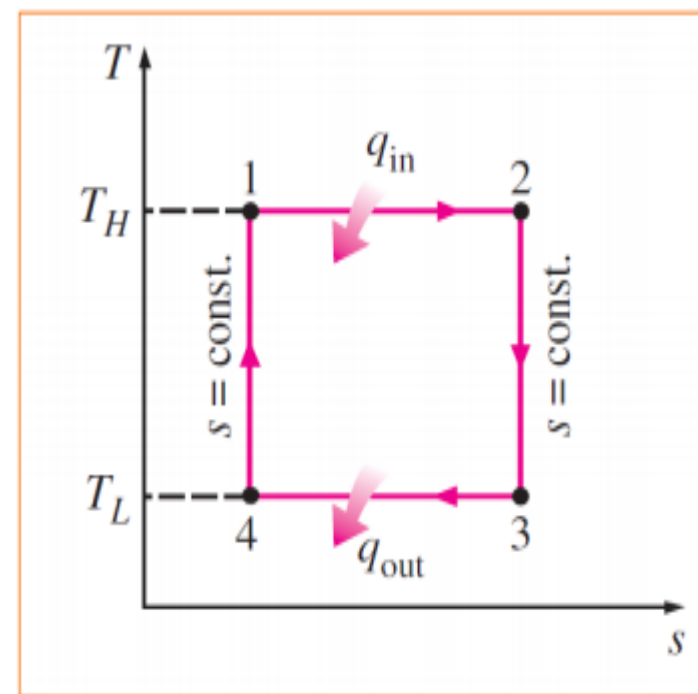
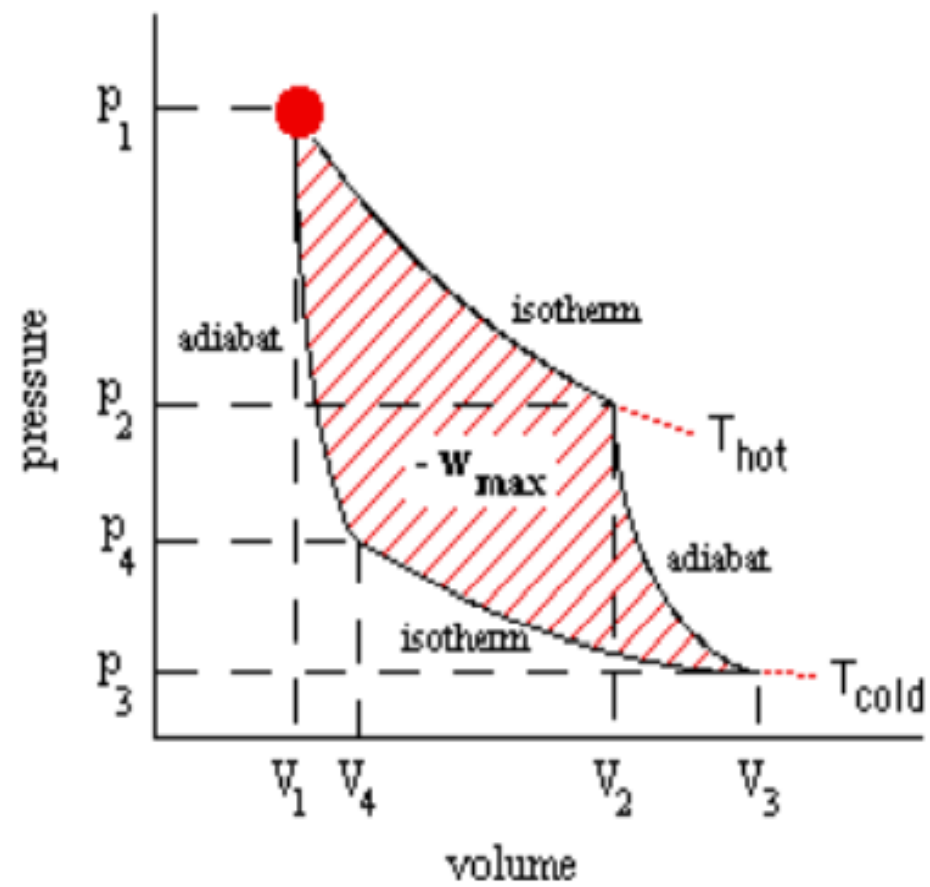
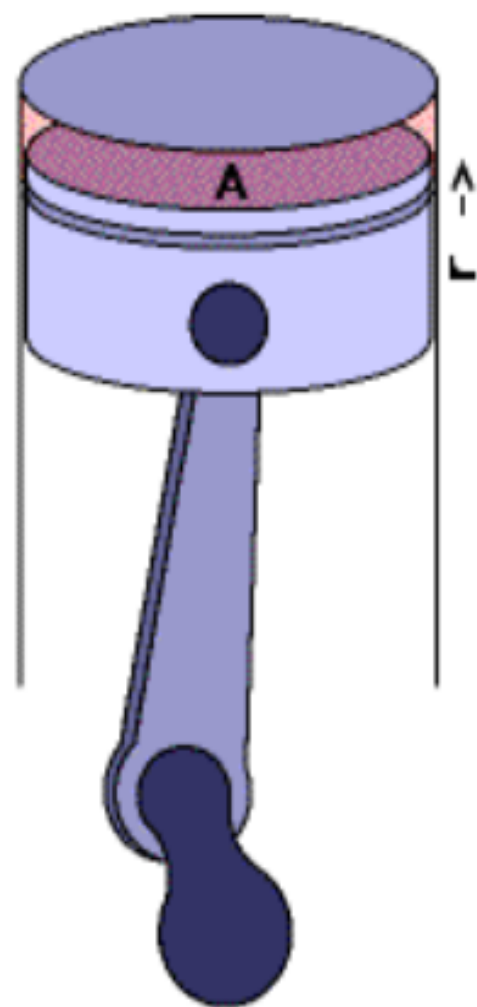
Carnot Cycle

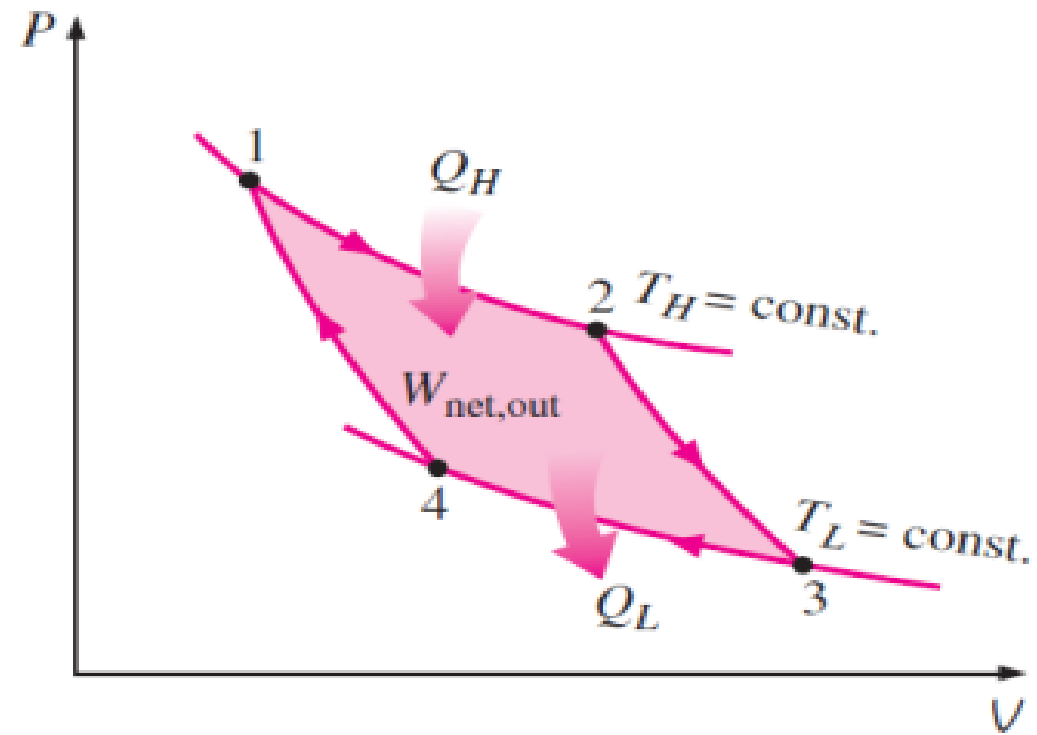
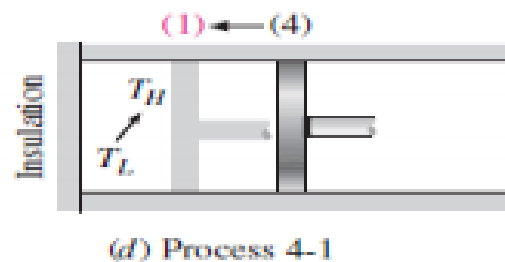
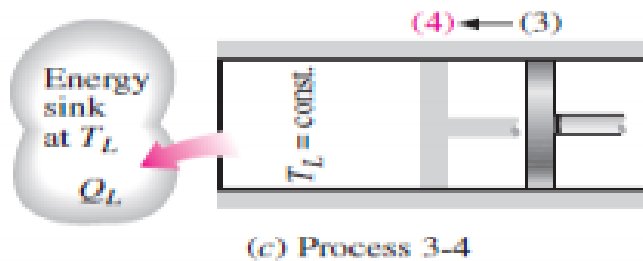
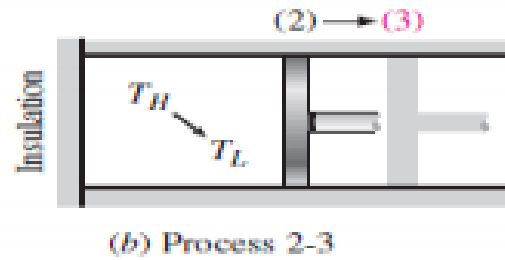
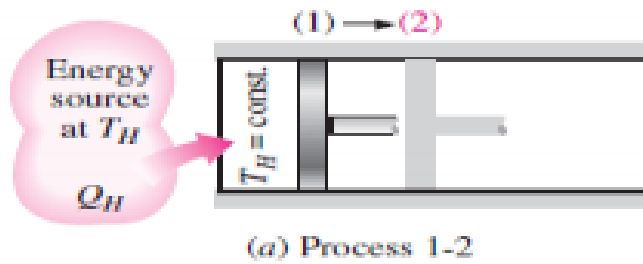


Sadi Carnot 1796-1832



- Probably the best known reversible cycle is the Carnot cycle, first proposed in 1824 by French engineer **Sadi Carnot**. The theoretical heat engine that operates on the Carnot cycle is called the **Carnot heat engine**. The Carnot cycle is composed of four reversible processes—two isothermal and two adiabatic

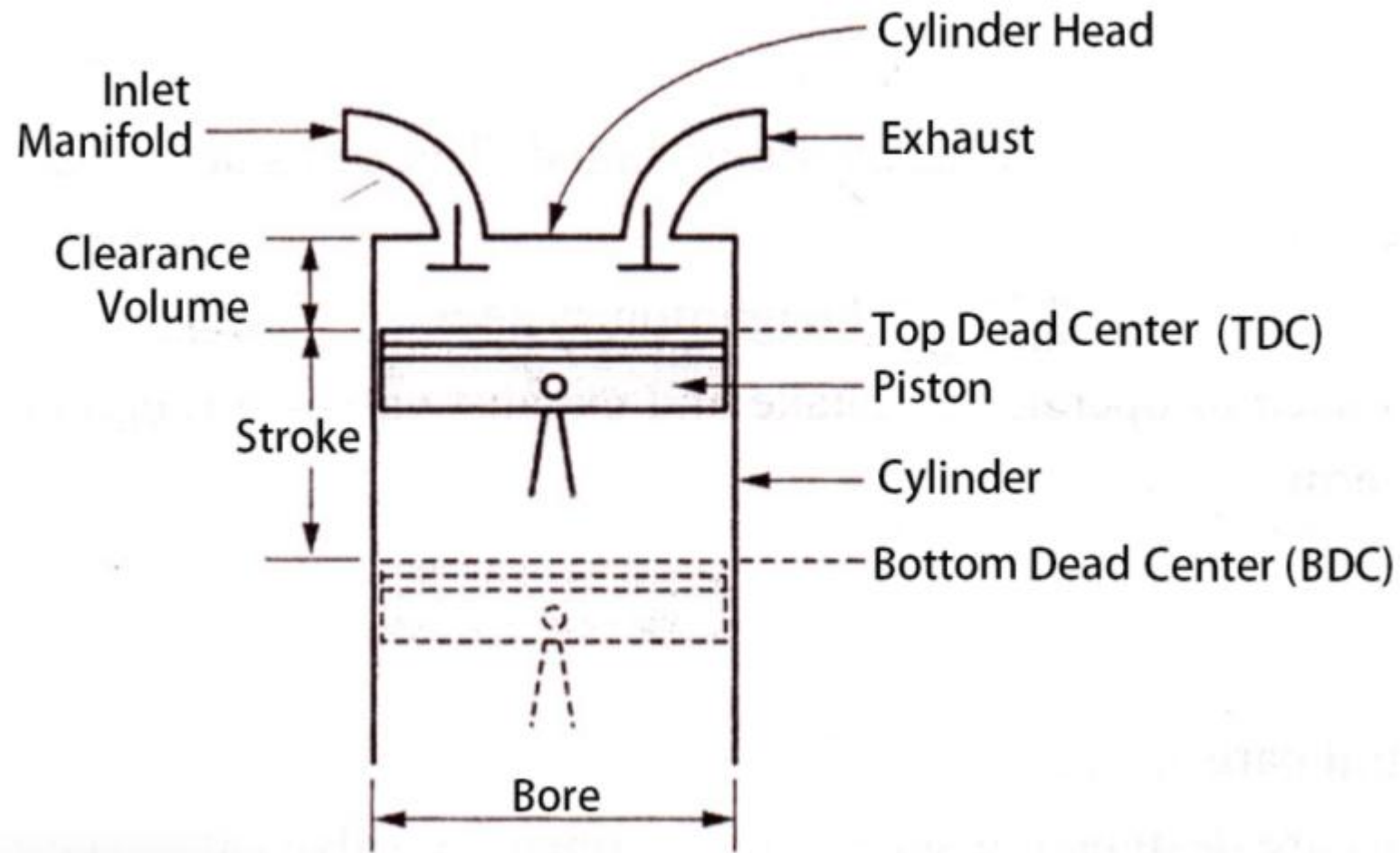




- Process 1-2, Isothermal expansion process
- Process 2-3, Isentropic expansion process
- Process 3-4, Isothermal compression process
- Process 4-1, Isentropic compression process

Carnot Cycle

- Area under curve 1-2-3 is the work done by the gas during the expansion part of the cycle, and the area under curve 3-4-1 is the work done on the gas during the compression part of the cycle.
- The area enclosed by the path of the cycle (area 1-2-3-4-1) is the difference between these two and represents the net work done during the cycle
- $V_3/V_2 = V_4/V_1$ or $V_3/V_4 = V_2/V_1$
- **Heat Supplied, $Q_H = P_1 V_1 \log_e(V_2/V_1)$**
- **Heat Rejected, $Q_L = P_3 V_3 \log_e(V_3/V_4)$**
- **Work Done, $W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}}$**
- **Thermal efficiency, $\eta = 1 - (Q_{\text{out}}/Q_{\text{in}})$**
 $= 1 - (Q_L / Q_H)$
 $= 1 - (T_L/T_H)$
 $= 1 - (\text{Low Temperature/High temperature})$



Equation for thermal efficiency

Heat added to air during the constant temperature process 1-2,

$$\begin{aligned} Q_s &= p_1 V_1 \ln \left(\frac{V_2}{V_1} \right) \\ &= m R T_1 \ln \left(\frac{V_2}{V_1} \right) \quad (\because pV = mRT) \end{aligned}$$

Heat transfer during adiabatic process 2-3 = 0

Heat rejected from air during constant temperature process 3-4,

$$\begin{aligned} Q_R &= p_3 V_3 \ln \left(\frac{V_4}{V_3} \right) \\ &= m R T_3 \ln \left(\frac{V_4}{V_3} \right) \end{aligned}$$

Heat transfer during adiabatic process, 4-1 = 0

Air standard efficiency (or thermal efficiency) is the ratio of net work done by the system to the heat supplied to the system.

$$\eta = \frac{W_{net}}{Q_s}$$

$$\eta = \frac{Q_S - Q_R}{Q_S} = 1 - \frac{Q_R}{Q_S}$$

$$= 1 - \frac{m R T_3 \ln (V_3/V_4)}{m R T_1 \ln (V_2/V_1)} \quad \text{————— (1)}$$

For the isentropic process 2-3, we can write,

$$\frac{T_2}{T_3} = \left(\frac{V_3}{V_2} \right)^{\gamma-1} \quad \text{————— (2)}$$

Since 1 - 2 and 3 - 4 are isothermal processes, we have, $T_1 = T_2$ and $T_3 = T_4$. Therefore, we can write

$$\frac{T_1}{T_4} = \left(\frac{V_3}{V_2} \right)^{\gamma-1} \quad \text{————— (3)}$$

For the isentropic process 4-1, we have

$$\frac{T_1}{T_4} = \left(\frac{V_4}{V_1} \right)^{\gamma-1}$$

$$\frac{V_3}{V_2} = \frac{V_1}{V_4} \quad \text{or} \quad \frac{V_3}{V_4} = \frac{V_2}{V_1} \quad \text{————— (4)}$$

Substituting Eq. (4) in Eq. (1), we get

$$\eta = 1 - \frac{m R T_3 \ln(V_3/V_4)}{m R T_1 \ln(V_3/V_4)}$$

$$= 1 - \frac{T_3}{T_1}$$

$$\eta = \frac{T_1 - T_3}{T_1}$$

Carnot cycle is not a practical cycle

- In order to achieve isothermal process, the piston has to move very slowly, so that the temperature remains constant during heat exchange. But, in order to achieve adiabatic process, the piston has to move very fast, so that there is no heat exchange with the surroundings. Since isothermal and adiabatic processes are to take place simultaneously, the cycle is not practically possible